

Strong Years program

Strong at 70



Build the leg, arm and grip strength daily life asks for:
chairs, stairs, the floor, the groceries.

12 weeks · four levels · every session scripted.

Chang Yin & Sun Yoon

Chang Yin and Sun Yoon are AI characters created by a team of people. Their life story is fiction. The exercises and recipes are built on published research for older adults (sources in the back). This is education, not medical advice. Check with your doctor before starting new exercise.

Strong at 70: how it works

Chang Yin

I'm Chang. AI character. Twelve weeks, one program, done properly. Build the leg, arm and grip strength daily life asks for: chairs, stairs, the floor, the groceries.

A little more, slowly. That's the whole method.

Who it's for. Adults 60+ who can stand from a chair (hands allowed) and want structured strength training at home. Every level from Rebuild to Iron.

Ask your doctor first if: Chest pain or unusual breathlessness with activity, fainting, recent surgery or fracture, a heart or kidney condition with symptoms, or a new hip or knee (follow your surgeon's precautions; kneeling may be off-limits).

What you need. Sturdy chair against a wall, kitchen counter, stairs with a rail, water jugs or grocery bags, a folded towel. Optional: the Strong Years band kit.

Every day: Stop if you feel chest pain or pressure, dizziness, unusual shortness of breath, or sharp pain. Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now. Pain rule (E41): Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist.

Your week

Mon: program session A · Tue: mobility · Wed: balance · Thu: program session B · Fri: breath + qigong · Sat: walk-and-talk · Sun: rest + stretch. Program sessions are marked in **bold** in the plan; the other days use your Strong Years Daily Practice sessions (DP numbers).

How it progresses

Three 4-week phases: **Foundation** (weeks 1–4, A1/B1), **Build** (weeks 5–8, A2/B2) and **Strengthen** (weeks 9–12, A3/B3).

Week in each phase	Rule
1st	Doses as written in the session. Learn the moves. Stop each set with one or two good reps left in the tank.
2nd	Add 1–2 reps per set (never past 12). Holds up 5 seconds (never past 30).
3rd	Where every rep felt easy two sessions in a row, use the Harder version or a heavier jug. Otherwise repeat week 2.
4th (easy week)	Easy week: same moves, one set fewer. Then do your checkpoint at the weekend.

Levels. Every session has four versions: Rebuild (chair-based), Steady, Strong and Iron. Move up only when every rep felt easy two sessions in a row. Move down any day. **Advanced** doses are never a starting point.

The 12-week plan

Week	Phase	Mon	Tue	Wed	Thu	Fri	Sat	Sun
1	Foundation	S70-A1	DP02	DP03	S70-B1	DP05	DP06	DP07
2	Foundation	S70-A1	DP09	DP10	S70-B1	DP12	DP13	DP14
3	Foundation	S70-A1	DP16	DP17	S70-B1	DP19	DP20	DP21
4 ★	Foundation	S70-A1	DP23	DP24	S70-B1	DP26	DP27	DP28
5	Build	S70-A2	DP30	DP03	S70-B2	DP05	DP06	DP07
6	Build	S70-A2	DP02	DP10	S70-B2	DP12	DP13	DP14
7	Build	S70-A2	DP09	DP17	S70-B2	DP19	DP20	DP21
8 ★	Build	S70-A2	DP16	DP24	S70-B2	DP26	DP27	DP28
9	Strengthen	S70-A3	DP23	DP03	S70-B3	DP05	DP06	DP07
10	Strengthen	S70-A3	DP30	DP10	S70-B3	DP12	DP13	DP14
11	Strengthen	S70-A3	DP02	DP17	S70-B3	DP19	DP20	DP21
12 ★	Strengthen	S70-A3	DP09	DP24	S70-B3	DP26	DP27	DP28

★ = easy week, then checkpoint. **Bold** = program session. DP = Daily Practice session number (sessions.json).

Red flags: Not for exercise. For your doctor, today: chest pain or pressure; fainting; sudden weakness or numbness in the face, arm or leg; trouble speaking; a sudden severe headache; new confusion; one calf that is swollen, red and painful; a hot, swollen joint; night pain that wakes you and doesn't change with position; fever with back pain; loss of bladder or bowel control or numbness in the saddle area; a fall where you hit your head (especially on blood thinners). Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now.

Your checkpoints

Start, then the end of weeks 4, 8 and 12. Same chair, same shoes, same time of day. These are fitness checks from published research, **not medical tests**.

Checkpoint	How	Start	Week 4	Week 8	Week 12
30-second chair stand (E11, E49)	Full stands in 30 s, arms crossed (H if hands used)				
Five chair stands, timed (E06)	Seconds for 5 stands, arms crossed. More than 15 s is one of the European screening signs of low muscle strength: mention it to your doctor				
30-second arm curl (E49)	Full curls, same weight each time				
Floor level (E45, E08)	0 = chair stands only · 1 = half-kneel and back with the chair · 2 = kneel and rise with the chair · 3 = sit to the floor and rise with the chair (someone home)				
Two-bag carry (E01)	Seconds carrying two bags or jugs along the counter, up to 60 s				

The program sessions

Full follow-along scripts for the six program sessions. The same content is in programs.json for the app and video pipeline.

S70-A1 · Phase 1 (Foundation): Legs and push

Type	Length	Set	Wardrobe	Voices
Strong at 70 program	10:35	SET-GARAGE	C-TRAIN-A	Chang

On screen the whole session: the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

Gentle-day swaps (the sore knee / sore back / low energy button): **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

Run of show

Time	Segment	Rebuild	Steady	Strong	Iron
0:00–0:35	Welcome	All levels together			
0:35–1:00	Ankle pumps Ankle pumps and circles	10 pumps	10 pumps	10 pumps	10 pumps
1:00–1:40	March Standing march at the counter	<i>Seated march:</i> 40 seconds	40 seconds	40 seconds	40 seconds

Time	Segment	Rebuild	Steady	Strong	Iron
1:40–2:00	Shoulder rolls	6 rolls	8 rolls	8 rolls	8 rolls + circles
2:00–3:50	Legs Chair sit-to-stand	2 sets of 8, hands on thighs	3 sets of 8	3 sets of 10, 3-second lower	3 sets of 10, jug at chest
3:50–5:20	Push Wall push-up	<i>Seated chest press:</i> 2 sets of 10	3 sets of 8	<i>Counter push-up:</i> 3 sets of 8	<i>Counter push-up:</i> 3 sets of 10, 3-second lower
5:20–6:40	Suitcase lift Suitcase lift (one-hand hinge)	<i>Seated hip hinge:</i> 2 sets of 8	2 sets of 8 each side	3 sets of 8 each side	3 sets of 8 each side, heavy bag
6:40–7:30	Heel raises Heel raises at the counter	2 sets of 12, seated	2 sets of 10	2 sets of 12, one hand	2 sets of 8 each leg
7:30–8:50	Floor practice Floor practice with a chair (kneel and rise)	<i>Chair sit-to-stand:</i> 1 set of 5 (no floor)	2 half-kneels each side	3 half-kneels each side	2 kneel-and-rise
8:50–9:30	Calf stretch Calf stretch at the wall	20 seconds each leg	20 seconds each leg	20 seconds each leg	20 seconds each leg
9:30–10:10	Cyclic sighing	30 seconds	40 seconds	40 seconds	40 seconds
10:10–10:35	Close	All levels together			

Script

0:00–0:35 • Welcome

Time	Who	Spoken	On screen
0:00	Chang	Strong at 70, phase one. Legs and push.	Strong at 70 • A1
0:10	Chang	Weeks one to four we learn the moves and own them.	Stop if chest pain, dizziness or sharp pain
0:17	Chang	Chair against the wall. Counter close.	
0:24	Chang	Chest pain, dizziness or sharp pain: stop, sit, and call your doctor.	

0:35–1:00 • Ankle pumps

Time	Who	Spoken	On screen
0:35	Chang	Sit tall. Chair against the wall.	Ankle pumps
0:42	Chang	Point your toes away, then pull them toward you. That's one pump.	Easier: Smaller movements
0:46	Chang	Big, slow movements.	
0:50	Chang	Feel your calves work.	
0:54	Chang	Breathe slowly.	

Cues: Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

Safety callouts: Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

1:00–1:40 • March

Time	Who	Spoken	On screen
1:00	Chang	Stand slowly. Hand on the counter.	March
1:19	Chang	March. Easy rhythm. Breathe steady.	Easier: Sit and march

Cues: Stand tall. Knee up, foot down softly. **Easier:** Sit and do the seated march. **Harder:** Lift knees higher, march faster, or let one hand hover above the counter.

1:40–2:00 • Shoulder rolls

Time	Who	Spoken	On screen
1:40	Chang	Sit tall. Chair against the wall.	Shoulder rolls
1:46	Chang	Lift your shoulders up toward your ears.	Easier: Just shrug and let go
1:49	Chang	Big, slow circles, backward.	
1:52	Chang	Let your arms hang heavy.	
1:55	Chang	Breathe in as you lift, out as you roll down.	

Cues: Big, slow circles, backward. Let your arms hang heavy. **Easier:** Just shrug up and let go. **Harder:** Add arm circles: small circles out to the sides, forward and back.

2:00–3:50 • Legs

Time	Who	Spoken	On screen
2:00	Chang	Hand on the counter. Chair behind you.	Legs
2:20	Chang	Cross your arms over your chest, or rest your hands on your thighs.	Easier: Higher seat, hands on thighs

Time	Who	Spoken	On screen
2:38	Chang	Nose over toes.	
2:49	Chang	One... two... breathe out... three... slow down... four.	
2:55	Chang	Knees point the same way as your toes. Don't let them cave inward.	
3:13	Chang	Breathe out as you stand. Breathe in as you sit.	

Cues: Nose over toes. Knees point the same way as your toes. Don't let them cave inward. Sit down slow. Two or three seconds. **Easier:** Put a firm cushion or folded blanket on the seat to make it higher, or push up with your hands on your thighs or on chair arms. **Harder:** Arms crossed, lower for a slow count of three, and pause one second just above the seat before touching down. Later: hold a gallon jug of water at your chest.

3:50–5:20 • Push

Time	Who	Spoken	On screen
3:50	Chang	Wall in front of you or behind you. Chair nearby.	Push
4:07	Chang	Bend your elbows and bring your chest toward the wall.	Easier: Closer to wall, half range
4:21	Chang	Body in one straight line, head to heels.	
4:30	Chang	One... two... breathe out... three... slow down... four.	
4:36	Chang	Chest to the wall, not your chin.	
4:50	Chang	Breathe out as you push away.	

Cues: Body in one straight line, head to heels. Chest to the wall, not your chin. **Easier:** Stand closer to the wall and move only halfway. **Harder:** Step your feet farther back. When 15 are easy, move to counter push-ups.

Safety callouts: Wrist pain? Make fists or hold the counter edge instead of flat palms.

5:20–6:40 • Suitcase lift

Time	Who	Spoken	On screen
5:20	Chang	Hand on the counter. Chair behind you.	Suitcase lift
5:35	Chang	Push your hips back with a long back and bend your knees a little.	Easier: Lighter jug, higher surface
5:48	Chang	Hips back. Long back.	
5:56	Chang	One... two... breathe out... three... slow down... four.	
6:01	Chang	Don't lean toward the jug. Stay square.	
6:14	Chang	Breathe out as you stand.	

Cues: Hips back. Long back. Don't lean toward the jug. Stay square.

Easier: A lighter jug, and lift it from a higher surface like the counter's edge of a table. **Harder:** A heavier bag, or lift from the bottom step instead of the chair.

Safety callouts: Osteoporosis: this is the safe bending pattern. Keep the back long, never curl. Osteoporosis or a past spine fracture: long back always, never round to reach.

6:40–7:30 • Heel raises

Time	Who	Spoken	On screen
6:40	Chang	Hand on the counter. Chair behind you.	Heel raises
6:51	Chang	Rise up onto the balls of your feet, as high as is comfortable.	Easier: Seated, or rise halfway
6:59	Chang	Tall like someone is pulling a string on top of your head.	
7:02	Chang	One... two... breathe out... three... slow down... four.	
7:07	Chang	Weight over the big toe and second toe, not rolling out.	
7:15	Chang	Breathe out as you rise.	

Cues: Tall like someone is pulling a string on top of your head. Weight over the big toe and second toe, not rolling out. **Easier:** Sit in your chair and raise your heels from there, or hold the counter with both hands and rise only halfway. **Harder:** One hand on the counter. Later: one leg at a time, with both hands on the counter.

Safety callouts: Calf cramps are common at first. Stretch and walk them out. One calf that is swollen, red and painful is not a cramp: that is for your doctor, today.

7:30–8:50 • Floor practice

Time	Who	Spoken	On screen
7:30	Chang	Chair or sofa beside you. Towel down. Someone else home.	Floor practice
7:54	Chang	Step back, lower one knee to the towel. Hands on the chair.	Easier: Half-kneel only, or chair stands

Time	Who	Spoken	On screen
8:15	Chang	Push up through the front foot. Breathe out. Stand still a moment.	Only with a sturdy chair and someone home

Cues: Hands on the chair the whole time. Slow down. Up is a push, not a jump. **Easier:** Only the first step: lower into a half-kneel on the towel and come back up, holding the chair. Or skip the floor and do sit-to-stands. **Harder:** Iron: from kneeling, sit down onto the floor beside the chair, then roll to your knees and rise with the chair. Only with someone home.

Safety callouts: Only try the floor with a sturdy chair or sofa next to you and someone home. Recent hip or knee replacement, or severe knee arthritis: follow your surgeon's or physical therapist's guidance first. New knee? Kneeling may not be allowed yet. Ask your surgeon. Sit a moment before you stand, then stand slowly while holding something steady.

8:50–9:30 • Calf stretch

Time	Who	Spoken	On screen
8:50	Chang	Hand on the counter. Chair behind you.	Calf stretch
8:59	Chang	Bend the front knee and lean gently toward the wall until you feel a stretch in the back calf.	Easier: Smaller step back
9:05	Chang	Back heel stays down.	
9:12	Chang	Toes point forward.	
9:18	Chang	Breathe slowly through the stretch.	

Cues: Back heel stays down. Toes point forward. **Easier:** Smaller step back. **Harder:** Bigger step back, or bend the back knee slightly to reach the lower calf.

9:30–10:10 • Cyclic sighing

Time	Who	Spoken	On screen
9:30	Chang	Sit. Two sniffs in through the nose. One long sigh out.	Cyclic sighing
9:49	Chang	No holding.	Easier: Just slow nose breathing

Cues: Two sniffs in, one long sigh out. No holding at any point. **Easier:** Just breathe slowly through your nose, making the out-breath a little longer. **Harder:** Build up to 5 minutes.

Safety callouts: If you feel light-headed, return to normal breathing.

10:10–10:35 • Close

Time	Who	Spoken	On screen
10:10	Chang	Legs first. Push second. Deposit made.	
10:23	Chang	Thursday: pull and carry.	

S70-B1 • Phase 1 (Foundation): Pull and carry

Type	Length	Set	Wardrobe	Voices
Strong at 70 program	10:50	SET-GARAGE	C-TRAIN-A	Chang

On screen the whole session: the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

Gentle-day swaps (the sore knee / sore back / low energy button): **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

Run of show

Time	Segment	Rebuild	Steady	Strong	Iron
0:00–0:35	Welcome	All levels together			
0:35–1:00	Ankle pumps Ankle pumps and circles	10 pumps	10 pumps	10 pumps	10 pumps

Time	Segment	Rebuild	Steady	Strong	Iron
1:00–1:40	March Standing march at the counter	<i>Seated march:</i> 40 seconds	40 seconds	40 seconds	40 seconds
1:40–2:00	Shoulder rolls	6 rolls	8 rolls	8 rolls	8 rolls + circles
2:00–3:30	Row One-arm jug row at the counter	<i>Seated towel pull-apart:</i> 4 pulls, 3-second squeeze	2 sets of 10 each arm	<i>Seated band row:</i> 3 sets of 10	<i>Seated band row:</i> 3 sets of 12, pause
3:30–5:00	Step-ups Step-ups on the bottom stair	<i>Seated march:</i> 2 × 30 seconds, knees high	2 sets of 10 toe taps	2 sets of 8 each leg	3 sets of 8 each leg, bag
5:00–6:20	Hinge Counter hip hinge	<i>Seated hip hinge:</i> 2 sets of 8	2 sets of 10	3 sets of 8, jug from the chair	3 sets of 8, bag from the step
6:20–7:15	Curls Water-bottle curl	2 sets of 10, bottles	2 sets of 10, half-gallon jug	3 sets of 10, half-gallon jug	3 sets of 10, gallon jug
7:15–8:25	One-side carry	<i>Towel wring (grip):</i> 3 wrings each way	2 × 20 seconds each side	2 × 30 seconds each side	3 × 30 seconds each side
8:25–9:05	Towel wring Towel wring (grip)	2 each way	3 each way	3 each way	3 each way, damp towel
9:05–9:45	Calf stretch Calf stretch at the wall	20 seconds each leg	20 seconds each leg	20 seconds each leg	20 seconds each leg
9:45–10:25	Cyclic sighing	30 seconds	40 seconds	40 seconds	40 seconds

Time	Segment	Rebuild	Steady	Strong	Iron
10:25–10:50	Close	All levels together			

Script

0:00–0:35 • Welcome

Time	Who	Spoken	On screen
0:00	Chang	Strong at 70. Pull and carry.	Strong at 70 • B1
0:10	Chang	Your back and your grip do the work you don't notice.	Stop if chest pain, dizziness or sharp pain
0:17	Chang	Bags, jugs, the band if you have it.	
0:24	Chang	Chest pain, dizziness or sharp pain: stop, sit, and call your doctor.	

0:35–1:00 • Ankle pumps

Time	Who	Spoken	On screen
0:35	Chang	Sit tall. Chair against the wall.	Ankle pumps
0:42	Chang	Point your toes away, then pull them toward you. That's one pump.	Easier: Smaller movements
0:46	Chang	Big, slow movements.	
0:50	Chang	Feel your calves work.	
0:54	Chang	Breathe slowly.	

Cues: Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

Safety callouts: Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

1:00–1:40 • March

Time	Who	Spoken	On screen
1:00	Chang	Stand slowly. Hand on the counter.	March
1:19	Chang	March. Easy rhythm. Breathe steady.	Easier: Sit and march

Cues: Stand tall. Knee up, foot down softly. **Easier:** Sit and do the seated march. **Harder:** Lift knees higher, march faster, or let one hand hover above the counter.

1:40–2:00 • Shoulder rolls

Time	Who	Spoken	On screen
1:40	Chang	Sit tall. Chair against the wall.	Shoulder rolls
1:46	Chang	Lift your shoulders up toward your ears.	Easier: Just shrug and let go
1:49	Chang	Big, slow circles, backward.	
1:52	Chang	Let your arms hang heavy.	
1:55	Chang	Breathe in as you lift, out as you roll down.	

Cues: Big, slow circles, backward. Let your arms hang heavy. **Easier:** Just shrug up and let go. **Harder:** Add arm circles: small circles out to the sides, forward and back.

2:00–3:30 • Row

Time	Who	Spoken	On screen
2:00	Chang	Hand on the counter. Chair behind you.	Row
2:17	Chang	Pull the jug up toward your ribs, elbow close to your body.	Easier: Towel pull or lighter jug
2:31	Chang	Pull with your elbow, not your hand.	
2:40	Chang	One... two... breathe out... three... slow down... four.	
2:46	Chang	Long back. Don't twist.	
3:00	Chang	Breathe out as you pull.	

Cues: Pull with your elbow, not your hand. Long back. Don't twist. **Easier:** Stand upright and pull a towel looped around a sturdy door handle (see Towel row), or use a lighter jug. **Harder:** Full gallon jug (about 8 pounds), or a bag with a few cans. Pause for one second at the top.

Safety callouts: Osteoporosis: keep the hinge small and the back long. Check the band for nicks before every use. Door must open away from you and be locked.

3:30–5:00 • Step-ups

Time	Who	Spoken	On screen
3:30	Chang	Hand on the rail.	Step-ups
3:47	Chang	Step up with your right foot, placing the whole foot on the step.	Easier: Toe taps, hand on rail
4:01	Chang	Whole foot on the step, not just the toes.	
4:10	Chang	One... two... breathe out... three... slow down... four.	

Time	Who	Spoken	On screen
4:16	Chang	Push through the heel of the top foot.	
4:30	Chang	Breathe out as you step up.	

Cues: Whole foot on the step, not just the toes. Push through the heel of the top foot. Rail hand stays on the rail. **Easier:** Toe taps: tap the step with one foot, then the other, holding the rail. Or use a lower step, like a sturdy doorway threshold. **Harder:** Step up without pushing off the bottom foot. Later: carry a light bag of groceries in the hand that isn't on the rail.

Safety callouts: If stairs make you dizzy or you have fallen on stairs, do toe taps only until you've practiced balance for two weeks.

5:00–6:20 • Hinge

Time	Who	Spoken	On screen
5:00	Chang	Hand on the counter. Chair behind you.	Hinge
5:15	Chang	Push your hips back, like closing a car door with your bottom.	Easier: Smaller hinge, hands stay down
5:28	Chang	Hips go back, not down.	
5:36	Chang	One... two... breathe out... three... slow down... four.	
5:41	Chang	Long back, chest proud. No rounding.	
5:54	Chang	Breathe out as you stand tall.	

Cues: Hips go back, not down. Long back, chest proud. No rounding. Stand up by squeezing your buttocks. **Easier:** Smaller movement, hands stay on the counter the whole time. **Harder:** Pick up a jug of water from the seat of a chair using the same hinge, and set it back down. Later: from a lower surface, like a step.

Safety callouts: Osteoporosis: this is the safe bending pattern. Keep the back long, never curl. Osteoporosis or a past spine fracture: this is the safe way to bend. Keep your back long; never round forward under load.

6:20–7:15 • Curls

Time	Who	Spoken	On screen
6:20	Chang	Sit tall. Chair against the wall.	Curls
6:31	Chang	Bend your elbows and bring the bottles toward your shoulders.	Easier: One arm, or empty bottles
6:40	Chang	Elbows stay by your sides.	
6:44	Chang	One... two... breathe out... three... slow down... four.	
6:49	Chang	No swinging.	
6:58	Chang	Breathe out as you curl up.	

Cues: Elbows stay by your sides. No swinging. **Easier:** One arm at a time, or empty bottles. **Harder:** Heavier: a half-gallon jug (about 4 pounds), then a full gallon jug (about 8 pounds), one at a time.

7:15–8:25 • One-side carry

Time	Who	Spoken	On screen
7:15	Chang	Clear path. Free hand trails along the counter.	One-side carry

Time	Who	Spoken	On screen
7:29	Chang	Stand tall. Don't lean away from the bag.	Easier: Lighter bag, hand on counter
7:40	Chang	Tall like a lamppost.	
7:51	Chang	Shoulders level.	
8:02	Chang	Breathe steadily. You should be able to talk.	

Cues: Tall like a lamppost. Shoulders level. **Easier:** A lighter bag, free hand on the counter the whole time. **Harder:** A heavier bag, or a longer walk.

Safety callouts: Hand arthritis flaring today? Squeeze gently, or skip and do the finger spread only. Pick up and put down with a hinge. Never round your back to lift.

8:25–9:05 • Towel wring

Time	Who	Spoken	On screen
8:25	Chang	Sit tall. Chair against the wall.	Towel wring
8:34	Chang	Wring the towel hard, like squeezing out water: right hand one way, left hand the other.	Easier: Squeeze a rolled sock
8:40	Chang	Squeeze from your whole hand, not just your fingertips.	
8:47	Chang	Shoulders relaxed.	
8:53	Chang	Breathe out as you squeeze.	

Cues: Squeeze from your whole hand, not just your fingertips. Shoulders relaxed. **Easier:** Squeeze a soft ball or rolled sock instead. **Harder:** Use a damp towel, or hold each wring for five seconds.

Safety callouts: Hand arthritis flaring today? Squeeze gently, or skip and do the finger spread only.

9:05–9:45 • Calf stretch

Time	Who	Spoken	On screen
9:05	Chang	Hand on the counter. Chair behind you.	Calf stretch
9:14	Chang	Bend the front knee and lean gently toward the wall until you feel a stretch in the back calf.	Easier: Smaller step back
9:20	Chang	Back heel stays down.	
9:27	Chang	Toes point forward.	
9:33	Chang	Breathe slowly through the stretch.	

Cues: Back heel stays down. Toes point forward. **Easier:** Smaller step back. **Harder:** Bigger step back, or bend the back knee slightly to reach the lower calf.

9:45–10:25 • Cyclic sighing

Time	Who	Spoken	On screen
9:45	Chang	Sit. Two sniffs in through the nose. One long sigh out.	Cyclic sighing
10:04	Chang	No holding.	Easier: Just slow nose breathing

Cues: Two sniffs in, one long sigh out. No holding at any point. **Easier:** Just breathe slowly through your nose, making the out-breath a little longer. **Harder:** Build up to 5 minutes.

Safety callouts: If you feel light-headed, return to normal breathing.

10:25–10:50 • Close

Time	Who	Spoken	On screen
10:25	Chang	Pull and carry. That's groceries, grandkids and doors.	
10:38	Chang	Monday: legs again.	

S70-A2 · Phase 2 (Build): Power and floor

Type	Length	Set	Wardrobe	Voices
Strong at 70 program	10:30	SET-GARAGE	C-TRAIN-A	Chang

On screen the whole session: the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

Gentle-day swaps (the sore knee / sore back / low energy button): **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

Run of show

Time	Segment	Rebuild	Steady	Strong	Iron
0:00–0:35	Welcome	All levels together			
0:35–1:00	Ankle pumps Ankle pumps and circles	10 pumps	10 pumps	10 pumps	10 pumps

Time	Segment	Rebuild	Steady	Strong	Iron
1:00–1:40	March Standing march at the counter	<i>Seated march:</i> 40 seconds	40 seconds	40 seconds	40 seconds
1:40–2:00	Shoulder rolls	6 rolls	8 rolls	8 rolls	8 rolls + circles
2:00–3:40	Power stands Power stand (fast up, slow down)	<i>Chair sit-to-stand:</i> 3 sets of 6, normal speed	3 sets of 5	3 sets of 6, jug	3 sets of 8, jug (advanced)
3:40–5:05	Push Counter push-up	<i>Seated chest press:</i> 3 sets of 10, band	2 sets of 8	3 sets of 10, 3-second lower	3 sets of 12, 3-second lower
5:05–6:35	Floor practice Floor practice with a chair (kneel and rise)	<i>Chair sit-to-stand:</i> 1 set of 6 (no floor)	2 kneel-and-rise	3 kneel-and-rise	2 sit-to-floor and rise
6:35–7:50	Suitcase lift Suitcase lift (one-hand hinge)	<i>Seated hip hinge:</i> 2 sets of 8, bottle at chest	3 sets of 8 each side	3 sets of 8 each side, from the step	3 sets of 10 each side, heavier
7:50–8:45	High wall sit	<i>Seated knee straightening:</i> 2 sets of 10 each leg	2 holds of 20 seconds	3 holds of 20 seconds	3 holds of 30 seconds
8:45–9:25	Calf stretch Calf stretch at the wall	20 seconds each leg	20 seconds each leg	20 seconds each leg	20 seconds each leg
9:25–10:05	Cyclic sighing	30 seconds	40 seconds	40 seconds	40 seconds

Time	Segment	Rebuild	Steady	Strong	Iron
10:05–10:30	Close	All levels together			

Script

0:00–0:35 • Welcome

Time	Who	Spoken	On screen
0:00	Chang	Phase two. Power and the floor.	Strong at 70 • A2
0:10	Chang	Fast up, slow down. Then practise the floor.	Stop if chest pain, dizziness or sharp pain
0:17	Chang	Chair against the wall. Someone home for the floor part.	
0:24	Chang	Chest pain, dizziness or sharp pain: stop, sit, and call your doctor.	

0:35–1:00 • Ankle pumps

Time	Who	Spoken	On screen
0:35	Chang	Sit tall. Chair against the wall.	Ankle pumps
0:42	Chang	Point your toes away, then pull them toward you. That's one pump.	Easier: Smaller movements
0:46	Chang	Big, slow movements.	
0:50	Chang	Feel your calves work.	
0:54	Chang	Breathe slowly.	

Cues: Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

Safety callouts: Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

1:00–1:40 • March

Time	Who	Spoken	On screen
1:00	Chang	Stand slowly. Hand on the counter.	March
1:19	Chang	March. Easy rhythm. Breathe steady.	Easier: Sit and march

Cues: Stand tall. Knee up, foot down softly. **Easier:** Sit and do the seated march. **Harder:** Lift knees higher, march faster, or let one hand hover above the counter.

1:40–2:00 • Shoulder rolls

Time	Who	Spoken	On screen
1:40	Chang	Sit tall. Chair against the wall.	Shoulder rolls
1:46	Chang	Lift your shoulders up toward your ears.	Easier: Just shrug and let go
1:49	Chang	Big, slow circles, backward.	
1:52	Chang	Let your arms hang heavy.	
1:55	Chang	Breathe in as you lift, out as you roll down.	

Cues: Big, slow circles, backward. Let your arms hang heavy. **Easier:** Just shrug up and let go. **Harder:** Add arm circles: small circles out to the sides, forward and back.

2:00–3:40 • Power stands

Time	Who	Spoken	On screen
2:00	Chang	Hand on the counter. Chair behind you.	Power stands
2:19	Chang	Sit tall, feet flat, arms crossed or hands on thighs.	Easier: Normal speed, hands on thighs
2:35	Chang	Fast up, slow down.	Advanced (Iron): not your starting point
2:45	Chang	One... two... breathe out... three... slow down... four.	
2:51	Chang	Quick is not the same as sloppy. Knees stay over toes.	
3:07	Chang	Breathe out hard as you stand.	

Cues: Fast up, slow down. Quick is not the same as sloppy. Knees stay over toes. **Easier:** Normal-speed sit-to-stand with hands on your thighs. **Harder:** Hold a jug of water at your chest, or pause one second just above the seat on the way down.

Safety callouts: Stop the fast part if you feel dizzy when you stand. Go back to normal speed.

3:40–5:05 • Push

Time	Who	Spoken	On screen
3:40	Chang	Hand on the counter. Chair behind you.	Push
3:56	Chang	Lower your chest toward the counter edge.	Easier: Wall push-ups
4:10	Chang	Straight line from head to heels.	

Time	Who	Spoken	On screen
4:18	Chang	One... two... breathe out... three... slow down... four.	
4:23	Chang	Squeeze your buttocks so your hips don't sag.	
4:37	Chang	Breathe out as you push up.	

Cues: Straight line from head to heels. Squeeze your buttocks so your hips don't sag. **Easier:** Wall push-ups. **Harder:** Lower slowly for three seconds. Later (Iron track): a sturdy low bench or bottom stair, only if the counter version is easy for 12 reps.

Safety callouts: Wrist pain? Make fists or hold the counter edge instead of flat palms.

5:05–6:35 • Floor practice

Time	Who	Spoken	On screen
5:05	Chang	Chair beside you. Towel down. Someone else home.	Floor practice
5:32	Chang	Kneel. Then front foot flat. Push up with hands on the chair.	Easier: Half-kneel only, or chair stands
5:56	Chang	Breathe out as you rise. Earn the floor.	Only with a sturdy chair and someone home

Cues: Hands on the chair the whole time. Slow down. Up is a push, not a jump. **Easier:** Only the first step: lower into a half-kneel on the towel and come back up, holding the chair. Or skip the floor and do sit-to-stands. **Harder:** Iron: from kneeling, sit down onto the floor beside the chair, then roll to your knees and rise with the chair. Only with someone home.

Safety callouts: Only try the floor with a sturdy chair or sofa next to you and someone home. Recent hip or knee replacement, or severe knee arthritis: follow your surgeon's or physical therapist's guidance first. New knee? Kneeling may not be allowed yet. Ask your surgeon. Sit a moment before you stand, then stand slowly while holding something steady.

6:35–7:50 • Suitcase lift

Time	Who	Spoken	On screen
6:35	Chang	Hand on the counter. Chair behind you.	Suitcase lift
6:50	Chang	Push your hips back with a long back and bend your knees a little.	Easier: Lighter jug, higher surface
7:02	Chang	Hips back. Long back.	
7:08	Chang	One... two... breathe out... three... slow down... four.	
7:14	Chang	Don't lean toward the jug. Stay square.	
7:26	Chang	Breathe out as you stand.	

Cues: Hips back. Long back. Don't lean toward the jug. Stay square.

Easier: A lighter jug, and lift it from a higher surface like the counter's edge of a table. **Harder:** A heavier bag, or lift from the bottom step instead of the chair.

Safety callouts: Osteoporosis: this is the safe bending pattern. Keep the back long, never curl. Osteoporosis or a past spine fracture: long back always, never round to reach.

7:50–8:45 • High wall sit

Time	Who	Spoken	On screen
7:50	Chang	Back on the wall. A few inches down. Chair beside you.	High wall sit
8:07	Chang	Keep talking. Count out loud.	Easier: Only 2 inches down, 10 s
8:22	Chang	High blood pressure? Short holds, and ask your doctor about longer ones.	Keep breathing: count out loud

Cues: Shallow is fine. This is not a contest of depth. Knees stay behind or over your toes. **Easier:** Slide down only 2 inches, or hold for just 10 seconds. **Harder:** Slide a little lower (never below where your knees are bent to a right angle), or hold up to 30 seconds.

Safety callouts: Breathe through it. High blood pressure or a heart condition? Ask your doctor before long holds.

8:45–9:25 • Calf stretch

Time	Who	Spoken	On screen
8:45	Chang	Hand on the counter. Chair behind you.	Calf stretch
8:54	Chang	Bend the front knee and lean gently toward the wall until you feel a stretch in the back calf.	Easier: Smaller step back
9:00	Chang	Back heel stays down.	
9:07	Chang	Toes point forward.	
9:13	Chang	Breathe slowly through the stretch.	

Cues: Back heel stays down. Toes point forward. **Easier:** Smaller step back. **Harder:** Bigger step back, or bend the back knee slightly to reach the lower calf.

9:25–10:05 • Cyclic sighing

Time	Who	Spoken	On screen
9:25	Chang	Sit. Two sniffs in through the nose. One long sigh out.	Cyclic sighing
9:44	Chang	No holding.	Easier: Just slow nose breathing

Cues: Two sniffs in, one long sigh out. No holding at any point. **Easier:** Just breathe slowly through your nose, making the out-breath a little longer. **Harder:** Build up to 5 minutes.

Safety callouts: If you feel light-headed, return to normal breathing.

10:05–10:30 • Close

Time	Who	Spoken	On screen
10:05	Chang	Getting up fast is what saves you when you trip.	
10:18	Chang	Thursday: stairs and carries.	

S70-B2 · Phase 2 (Build): Stairs and carries

Type	Length	Set	Wardrobe	Voices
Strong at 70 program	10:25	SET-STOOP	C-TRAIN-A	Chang

On screen the whole session: the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

Gentle-day swaps (the sore knee / sore back / low energy button): **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

Run of show

Time	Segment	Rebuild	Steady	Strong	Iron
0:00–0:35	Welcome	All levels together			
0:35–1:00	Ankle pumps Ankle pumps and circles	10 pumps	10 pumps	10 pumps	10 pumps
1:00–1:40	March Standing march at the counter	<i>Seated march:</i> 40 seconds	40 seconds	40 seconds	40 seconds

Time	Segment	Rebuild	Steady	Strong	Iron
1:40–2:00	Shoulder rolls	6 rolls	8 rolls	8 rolls	8 rolls + circles
2:00–3:40	Stair practice Stair practice with the rail	<i>Seated march:</i> 3 × 30 seconds, knees high	Bottom 3 steps, 4 times	1 flight, twice	2 flights, bag in free hand
3:40–5:00	Row Seated band row	2 sets of 10, close to the door	3 sets of 10	3 sets of 12, pause	3 sets of 12, stronger band
5:00–6:10	Two-hand carry Grocery carry	<i>Towel wring (grip):</i> 3 wrings each way	2 × 30 seconds	3 × 40 seconds	3 × 45 seconds, heavy
6:10–7:00	Band pull-apart	2 sets of 8	2 sets of 10	3 sets of 10	3 sets of 12
7:00–8:00	Overhead press Seated overhead press (to comfortable height)	2 sets of 6, one arm at a time	2 sets of 8	3 sets of 10	3 sets of 10, half-gallon jugs
8:00–8:40	Towel wring Towel wring (grip)	2 each way	3 each way	3 each way	3 each way, damp towel
8:40–9:20	Calf stretch Calf stretch at the wall	20 seconds each leg	20 seconds each leg	20 seconds each leg	20 seconds each leg
9:20–10:00	Cyclic sighing	30 seconds	40 seconds	40 seconds	40 seconds
10:00–10:25	Close	All levels together			

Script

0:00–0:35 • Welcome

Time	Who	Spoken	On screen
0:00	Chang	Phase two. Stairs and carries.	Strong at 70 • B2
0:10	Chang	Rail hand never lets go.	Stop if chest pain, dizziness or sharp pain
0:17	Chang	Bags ready.	
0:24	Chang	Chest pain, dizziness or sharp pain: stop, sit, and call your doctor.	

0:35–1:00 • Ankle pumps

Time	Who	Spoken	On screen
0:35	Chang	Sit tall. Chair against the wall.	Ankle pumps
0:42	Chang	Point your toes away, then pull them toward you. That's one pump.	Easier: Smaller movements
0:46	Chang	Big, slow movements.	
0:50	Chang	Feel your calves work.	
0:54	Chang	Breathe slowly.	

Cues: Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

Safety callouts: Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

1:00–1:40 • March

Time	Who	Spoken	On screen
1:00	Chang	Stand slowly. Hand on the counter.	March
1:19	Chang	March. Easy rhythm. Breathe steady.	Easier: Sit and march

Cues: Stand tall. Knee up, foot down softly. **Easier:** Sit and do the seated march. **Harder:** Lift knees higher, march faster, or let one hand hover above the counter.

1:40–2:00 • Shoulder rolls

Time	Who	Spoken	On screen
1:40	Chang	Sit tall. Chair against the wall.	Shoulder rolls
1:46	Chang	Lift your shoulders up toward your ears.	Easier: Just shrug and let go
1:49	Chang	Big, slow circles, backward.	
1:52	Chang	Let your arms hang heavy.	
1:55	Chang	Breathe in as you lift, out as you roll down.	

Cues: Big, slow circles, backward. Let your arms hang heavy. **Easier:** Just shrug up and let go. **Harder:** Add arm circles: small circles out to the sides, forward and back.

2:00–3:40 • Stair practice

Time	Who	Spoken	On screen
2:00	Chang	Hand on the rail.	Stair practice
2:19	Chang	Walk up one step at a time at a steady pace, whole foot on each step.	Easier: Bottom 3 steps only

Time	Who	Spoken	On screen
2:35	Chang	Whole foot on the step.	
2:51	Chang	Rail hand never lets go.	
3:07	Chang	Breathe steadily. Rest at the top as long as you like.	

Cues: Whole foot on the step. Rail hand never lets go. **Easier:** Just the bottom 3 steps, up and down. **Harder:** Two flights, or carry a light bag in the free hand.

Safety callouts: Chest pain or unusual breathlessness on stairs: stop, sit, and call your doctor.

3:40–5:00 • Row

Time	Who	Spoken	On screen
3:40	Chang	Sit or stand with the chair behind you. Check the band first.	Row
3:55	Chang	Pull your elbows back past your ribs.	Easier: Closer to the door, or towel
4:08	Chang	Elbows close to your body.	
4:16	Chang	One... two... breathe out... three... slow down... four.	
4:21	Chang	Tall chest, shoulders down.	
4:34	Chang	Breathe out as you pull.	

Cues: Elbows close to your body. Tall chest, shoulders down. **Easier:** Sit closer to the door so the band is slacker, or do the towel pull-apart.

Harder: Scoot back for more tension, or pause 2 seconds with elbows back.

Safety callouts: Check the band for nicks before every use. Door must open away from you and be locked.

5:00–6:10 • Two-hand carry

Time	Who	Spoken	On screen
5:00	Chang	Clear path. Walk along the counter or a hallway wall.	Two-hand carry
5:14	Chang	Stand tall, shoulders down.	Easier: One light bag, hand on counter
5:25	Chang	Tall, like you're proud of the groceries.	
5:36	Chang	Short, steady steps.	
5:47	Chang	Breathe steadily. You should be able to talk.	

Cues: Tall, like you're proud of the groceries. Short, steady steps. **Easier:** One light bag, and keep your free hand trailing along the counter. **Harder:** Heavier: two full gallon jugs (about 8 pounds each). Iron track: two heavier bags or kettlebells.

Safety callouts: Hand arthritis flaring today? Squeeze gently, or skip and do the finger spread only. Pick up and put down with a hinge. Never round your back to lift.

6:10–7:00 • Band pull-apart

Time	Who	Spoken	On screen
6:10	Chang	Sit or stand with the chair behind you. Check the band first.	Band pull-apart
6:21	Chang	Pull the band apart toward your chest by opening your arms.	Easier: Hands wider on the band

Time	Who	Spoken	On screen
6:29	Chang	Shoulders down.	
6:32	Chang	One... two... breathe out... three... slow down... four.	
6:37	Chang	Slow on the way back.	
6:45	Chang	Breathe out as you pull apart.	

Cues: Shoulders down. Slow on the way back. **Easier:** Hands wider apart on the band (less tension), or the towel pull-apart. **Harder:** Hands closer together, or a 2-second squeeze.

7:00–8:00 • Overhead press

Time	Who	Spoken	On screen
7:00	Chang	Sit tall. Chair against the wall.	Overhead press
7:12	Chang	Press the bottles up toward the ceiling, only as high as is comfortable.	Easier: One arm, press at a slant
7:22	Chang	Ribs down. Don't arch your back.	
7:27	Chang	One... two... breathe out... three... slow down... four.	
7:31	Chang	Only to comfortable height.	
7:41	Chang	Breathe out as you press up.	

Cues: Ribs down. Don't arch your back. Only to comfortable height. **Easier:** One arm at a time, or press forward and up at a slant instead of straight up. **Harder:** Heavier bottles or a half-gallon jug in each hand; stand at the counter to press.

Safety callouts: Shoulder pain or a known rotator cuff problem? Only lift to a comfortable height, or skip it.

8:00–8:40 • Towel wring

Time	Who	Spoken	On screen
8:00	Chang	Sit tall. Chair against the wall.	Towel wring
8:09	Chang	Wring the towel hard, like squeezing out water: right hand one way, left hand the other.	Easier: Squeeze a rolled sock
8:15	Chang	Squeeze from your whole hand, not just your fingertips.	
8:22	Chang	Shoulders relaxed.	
8:28	Chang	Breathe out as you squeeze.	

Cues: Squeeze from your whole hand, not just your fingertips. Shoulders relaxed. **Easier:** Squeeze a soft ball or rolled sock instead. **Harder:** Use a damp towel, or hold each wring for five seconds.

Safety callouts: Hand arthritis flaring today? Squeeze gently, or skip and do the finger spread only.

8:40–9:20 • Calf stretch

Time	Who	Spoken	On screen
8:40	Chang	Hand on the counter. Chair behind you.	Calf stretch
8:49	Chang	Bend the front knee and lean gently toward the wall until you feel a stretch in the back calf.	Easier: Smaller step back
8:55	Chang	Back heel stays down.	
9:02	Chang	Toes point forward.	

Time	Who	Spoken	On screen
9:08	Chang	Breathe slowly through the stretch.	

Cues: Back heel stays down. Toes point forward. **Easier:** Smaller step back. **Harder:** Bigger step back, or bend the back knee slightly to reach the lower calf.

9:20–10:00 • Cyclic sighing

Time	Who	Spoken	On screen
9:20	Chang	Sit. Two sniffs in through the nose. One long sigh out.	Cyclic sighing
9:39	Chang	No holding.	Easier: Just slow nose breathing

Cues: Two sniffs in, one long sigh out. No holding at any point. **Easier:** Just breathe slowly through your nose, making the out-breath a little longer. **Harder:** Build up to 5 minutes.

Safety callouts: If you feel light-headed, return to normal breathing.

10:00–10:25 • Close

Time	Who	Spoken	On screen
10:00	Chang	Stairs and bags. You trained real life today.	
10:13	Chang	Rest well.	

S70-A3 • Phase 3 (Strengthen): Strong

Type	Length	Set	Wardrobe	Voices
Strong at 70 program	10:30	SET-GARAGE	C-TRAIN-A	Chang

On screen the whole session: the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

Gentle-day swaps (the sore knee / sore back / low energy button): **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

Run of show

Time	Segment	Rebuild	Steady	Strong	Iron
0:00–0:35	Welcome	All levels together			
0:35–1:00	Ankle pumps Ankle pumps and circles	10 pumps	10 pumps	10 pumps	10 pumps

Time	Segment	Rebuild	Steady	Strong	Iron
1:00–1:40	March Standing march at the counter	<i>Seated march:</i> 40 seconds	40 seconds	40 seconds	40 seconds
1:40–2:00	Shoulder rolls	6 rolls	8 rolls	8 rolls	8 rolls + circles
2:00–3:50	Legs Chair sit-to-stand	3 sets of 8, arms crossed if you can	3 sets of 10, jug at chest	<i>Counter split squat (Iron track):</i> 3 sets of 8 each leg	<i>Counter split squat (Iron track):</i> 3 sets of 10 each leg, jug (advanced)
3:50–5:10	Push Counter push-up	<i>Seated chest press:</i> 3 sets of 12, band	3 sets of 10	3 sets of 12, 3-second lower	3 sets of 12, pause at the bottom
5:10–6:40	Floor practice Floor practice with a chair (kneel and rise)	<i>Chair sit-to-stand:</i> 2 sets of 6 (no floor)	3 kneel-and-rise	2 sit-to-floor and rise	3 sit-to-floor and rise
6:40–7:55	Suitcase lift Suitcase lift (one-hand hinge)	<i>Seated hip hinge:</i> 3 sets of 8, bottle at chest	3 sets of 8 each side, from the step	3 sets of 10 each side, heavier	3 sets of 12 each side, heavy bag
7:55–8:45	Heel raises Heel raises at the counter	3 sets of 12, seated	3 sets of 12, one hand	3 sets of 8 each leg	3 sets of 12 each leg
8:45–9:25	Calf stretch Calf stretch at the wall	20 seconds each leg	20 seconds each leg	20 seconds each leg	20 seconds each leg
9:25–10:05	Cyclic sighing	30 seconds	40 seconds	40 seconds	40 seconds

Time	Segment	Rebuild	Steady	Strong	Iron
10:05–10:30	Close	All levels together			

Script

0:00–0:35 • Welcome

Time	Who	Spoken	On screen
0:00	Chang	Phase three. Strong.	Strong at 70 • A3
0:10	Chang	Heavier where you can, slower where you can't.	Stop if chest pain, dizziness or sharp pain
0:17	Chang	Chair against the wall. Someone home for the floor.	
0:24	Chang	Chest pain, dizziness or sharp pain: stop, sit, and call your doctor.	

0:35–1:00 • Ankle pumps

Time	Who	Spoken	On screen
0:35	Chang	Sit tall. Chair against the wall.	Ankle pumps
0:42	Chang	Point your toes away, then pull them toward you. That's one pump.	Easier: Smaller movements
0:46	Chang	Big, slow movements.	
0:50	Chang	Feel your calves work.	
0:54	Chang	Breathe slowly.	

Cues: Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

Safety callouts: Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

1:00–1:40 • March

Time	Who	Spoken	On screen
1:00	Chang	Stand slowly. Hand on the counter.	March
1:19	Chang	March. Easy rhythm. Breathe steady.	Easier: Sit and march

Cues: Stand tall. Knee up, foot down softly. **Easier:** Sit and do the seated march. **Harder:** Lift knees higher, march faster, or let one hand hover above the counter.

1:40–2:00 • Shoulder rolls

Time	Who	Spoken	On screen
1:40	Chang	Sit tall. Chair against the wall.	Shoulder rolls
1:46	Chang	Lift your shoulders up toward your ears.	Easier: Just shrug and let go
1:49	Chang	Big, slow circles, backward.	
1:52	Chang	Let your arms hang heavy.	
1:55	Chang	Breathe in as you lift, out as you roll down.	

Cues: Big, slow circles, backward. Let your arms hang heavy. **Easier:** Just shrug up and let go. **Harder:** Add arm circles: small circles out to the sides, forward and back.

2:00–3:50 • Legs

Time	Who	Spoken	On screen
2:00	Chang	Hand on the counter. Chair behind you.	Legs
2:20	Chang	Cross your arms over your chest, or rest your hands on your thighs.	Easier: Higher seat, hands on thighs
2:38	Chang	Nose over toes.	Advanced (Iron): not your starting point
2:49	Chang	One... two... breathe out... three... slow down... four.	
2:55	Chang	Knees point the same way as your toes. Don't let them cave inward.	
3:13	Chang	Breathe out as you stand. Breathe in as you sit.	

Cues: Nose over toes. Knees point the same way as your toes. Don't let them cave inward. Sit down slow. Two or three seconds. **Easier:** Put a firm cushion or folded blanket on the seat to make it higher, or push up with your hands on your thighs or on chair arms. **Harder:** Arms crossed, lower for a slow count of three, and pause one second just above the seat before touching down. Later: hold a gallon jug of water at your chest.

Safety callouts: Advanced. Only if your knees feel good with step-ups.

3:50–5:10 • Push

Time	Who	Spoken	On screen
3:50	Chang	Hand on the counter. Chair behind you.	Push
4:05	Chang	Lower your chest toward the counter edge.	Easier: Wall push-ups

Time	Who	Spoken	On screen
4:18	Chang	Straight line from head to heels.	
4:26	Chang	One... two... breathe out... three... slow down... four.	
4:31	Chang	Squeeze your buttocks so your hips don't sag.	
4:44	Chang	Breathe out as you push up.	

Cues: Straight line from head to heels. Squeeze your buttocks so your hips don't sag. **Easier:** Wall push-ups. **Harder:** Lower slowly for three seconds. Later (Iron track): a sturdy low bench or bottom stair, only if the counter version is easy for 12 reps.

Safety callouts: Wrist pain? Make fists or hold the counter edge instead of flat palms.

5:10–6:40 • Floor practice

Time	Who	Spoken	On screen
5:10	Chang	Chair beside you. Towel down. Someone home.	Floor practice
5:37	Chang	Down with control. Up with a push. Breathe out.	Easier: Half-kneel only, or chair stands
6:01	Chang	Stand still a moment before you walk.	Only with a sturdy chair and someone home

Cues: Hands on the chair the whole time. Slow down. Up is a push, not a jump. **Easier:** Only the first step: lower into a half-kneel on the towel and come back up, holding the chair. Or skip the floor and do sit-to-stands. **Harder:** Iron: from kneeling, sit down onto the floor beside the chair, then roll to your knees and rise with the chair. Only with someone home.

Safety callouts: Only try the floor with a sturdy chair or sofa next to you and someone home. Recent hip or knee replacement, or severe knee arthritis: follow your surgeon's or physical therapist's guidance first. New knee? Kneeling may not be allowed yet. Ask your surgeon. Sit a moment before you stand, then stand slowly while holding something steady.

6:40–7:55 • Suitcase lift

Time	Who	Spoken	On screen
6:40	Chang	Hand on the counter. Chair behind you.	Suitcase lift
6:55	Chang	Push your hips back with a long back and bend your knees a little.	Easier: Lighter jug, higher surface
7:07	Chang	Hips back. Long back.	
7:13	Chang	One... two... breathe out... three... slow down... four.	
7:19	Chang	Don't lean toward the jug. Stay square.	
7:31	Chang	Breathe out as you stand.	

Cues: Hips back. Long back. Don't lean toward the jug. Stay square.

Easier: A lighter jug, and lift it from a higher surface like the counter's edge of a table. **Harder:** A heavier bag, or lift from the bottom step instead of the chair.

Safety callouts: Osteoporosis: this is the safe bending pattern. Keep the back long, never curl. Osteoporosis or a past spine fracture: long back always, never round to reach.

7:55–8:45 • Heel raises

Time	Who	Spoken	On screen
7:55	Chang	Hand on the counter. Chair behind you.	Heel raises
8:06	Chang	Rise up onto the balls of your feet, as high as is comfortable.	Easier: Seated, or rise halfway
8:14	Chang	Tall like someone is pulling a string on top of your head.	
8:17	Chang	One... two... breathe out... three... slow down... four.	
8:22	Chang	Weight over the big toe and second toe, not rolling out.	
8:30	Chang	Breathe out as you rise.	

Cues: Tall like someone is pulling a string on top of your head. Weight over the big toe and second toe, not rolling out. **Easier:** Sit in your chair and raise your heels from there, or hold the counter with both hands and rise only halfway. **Harder:** One hand on the counter. Later: one leg at a time, with both hands on the counter.

Safety callouts: Calf cramps are common at first. Stretch and walk them out. One calf that is swollen, red and painful is not a cramp: that is for your doctor, today.

8:45–9:25 • Calf stretch

Time	Who	Spoken	On screen
8:45	Chang	Hand on the counter. Chair behind you.	Calf stretch
8:54	Chang	Bend the front knee and lean gently toward the wall until you feel a stretch in the back calf.	Easier: Smaller step back
9:00	Chang	Back heel stays down.	
9:07	Chang	Toes point forward.	
9:13	Chang	Breathe slowly through the stretch.	

Cues: Back heel stays down. Toes point forward. **Easier:** Smaller step back. **Harder:** Bigger step back, or bend the back knee slightly to reach the lower calf.

9:25–10:05 • Cyclic sighing

Time	Who	Spoken	On screen
9:25	Chang	Sit. Two sniffs in through the nose. One long sigh out.	Cyclic sighing
9:44	Chang	No holding.	Easier: Just slow nose breathing

Cues: Two sniffs in, one long sigh out. No holding at any point. **Easier:** Just breathe slowly through your nose, making the out-breath a little longer. **Harder:** Build up to 5 minutes.

Safety callouts: If you feel light-headed, return to normal breathing.

10:05–10:30 • Close

Time	Who	Spoken	On screen
10:05	Chang	Twelve weeks ago, this was a lot. Look at you.	
10:18	Chang	Thursday: the last B session.	

S70-B3 • Phase 3 (Strengthen): Strong carries

Type	Length	Set	Wardrobe	Voices
Strong at 70 program	9:50	SET-STOOP	C-TRAIN-A	Chang

On screen the whole session: the "AI character" corner tag. In the first 30 seconds and before each hard block: "Stop if chest pain, dizziness or sharp pain."

Gentle-day swaps (the sore knee / sore back / low energy button): **Sore knee:** Use the Rebuild version of any standing leg move; skip floor practice and stairs today. **Sore back:** Seated versions of hinges and carries; keep every bend long-backed. **Low energy:** Do the warm-up, the first two blocks and the cool-down.

Run of show

Time	Segment	Rebuild	Steady	Strong	Iron
0:00–0:35	Welcome	All levels together			
0:35–1:00	Ankle pumps Ankle pumps and circles	10 pumps	10 pumps	10 pumps	10 pumps

Time	Segment	Rebuild	Steady	Strong	Iron
1:00–1:40	March Standing march at the counter	<i>Seated march:</i> 40 seconds	40 seconds	40 seconds	40 seconds
1:40–2:00	Shoulder rolls	6 rolls	8 rolls	8 rolls	8 rolls + circles
2:00–3:35	Step-ups Step-ups on the bottom stair	<i>Seated march:</i> 3 × 30 seconds, knees high	3 sets of 8 each leg	3 sets of 8 each leg, bag	3 sets of 10 each leg, heavier bag
3:35–4:55	Row Seated band row	3 sets of 10	3 sets of 12, pause	3 sets of 12, stronger band	<i>One-arm jug row at the counter:</i> 3 sets of 12 each arm, heavy bag
4:55–5:55	Band press-out Band press-out (anti-twist)	2 sets of 6 each side, seated	2 sets of 8 each side	3 sets of 8 each side, 5-second hold	3 sets of 10 each side, 5-second hold
5:55–7:10	Two-hand carry Grocery carry	<i>Towel wring (grip):</i> 4 wrings each way	3 × 30 seconds, heavier	3 × 45 seconds, heavy	3 × 60 seconds, heavy
7:10–8:05	Curls Water-bottle curl	2 sets of 10, half-gallon jug	3 sets of 10, half-gallon jug	3 sets of 10, gallon jug	3 sets of 12, gallon jug
8:05–8:45	Calf stretch Calf stretch at the wall	20 seconds each leg	20 seconds each leg	20 seconds each leg	20 seconds each leg
8:45–9:25	Cyclic sighing	30 seconds	40 seconds	40 seconds	40 seconds

Time	Segment	Rebuild	Steady	Strong	Iron
9:25–9:50	Close	All levels together			

Script

0:00–0:35 • Welcome

Time	Who	Spoken	On screen
0:00	Chang	Phase three. Strong carries, stairs, pulls.	Strong at 70 • B3
0:10	Chang	Rail hand never lets go.	Stop if chest pain, dizziness or sharp pain
0:17	Chang	Bags ready.	
0:24	Chang	Chest pain, dizziness or sharp pain: stop, sit, and call your doctor.	

0:35–1:00 • Ankle pumps

Time	Who	Spoken	On screen
0:35	Chang	Sit tall. Chair against the wall.	Ankle pumps
0:42	Chang	Point your toes away, then pull them toward you. That's one pump.	Easier: Smaller movements
0:46	Chang	Big, slow movements.	
0:50	Chang	Feel your calves work.	
0:54	Chang	Breathe slowly.	

Cues: Big, slow movements. Feel your calves work. **Easier:** Smaller movements. **Harder:** Lift one foot off the floor while you circle.

Safety callouts: Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

1:00–1:40 • March

Time	Who	Spoken	On screen
1:00	Chang	Stand slowly. Hand on the counter.	March
1:19	Chang	March. Easy rhythm. Breathe steady.	Easier: Sit and march

Cues: Stand tall. Knee up, foot down softly. **Easier:** Sit and do the seated march. **Harder:** Lift knees higher, march faster, or let one hand hover above the counter.

1:40–2:00 • Shoulder rolls

Time	Who	Spoken	On screen
1:40	Chang	Sit tall. Chair against the wall.	Shoulder rolls
1:46	Chang	Lift your shoulders up toward your ears.	Easier: Just shrug and let go
1:49	Chang	Big, slow circles, backward.	
1:52	Chang	Let your arms hang heavy.	
1:55	Chang	Breathe in as you lift, out as you roll down.	

Cues: Big, slow circles, backward. Let your arms hang heavy. **Easier:** Just shrug up and let go. **Harder:** Add arm circles: small circles out to the sides, forward and back.

2:00–3:35 • Step-ups

Time	Who	Spoken	On screen
2:00	Chang	Hand on the rail.	Step-ups
2:18	Chang	Step up with your right foot, placing the whole foot on the step.	Easier: Toe taps, hand on rail
2:33	Chang	Whole foot on the step, not just the toes.	
2:42	Chang	One... two... breathe out... three... slow down... four.	
2:48	Chang	Push through the heel of the top foot.	
3:03	Chang	Breathe out as you step up.	

Cues: Whole foot on the step, not just the toes. Push through the heel of the top foot. Rail hand stays on the rail. **Easier:** Toe taps: tap the step with one foot, then the other, holding the rail. Or use a lower step, like a sturdy doorway threshold. **Harder:** Step up without pushing off the bottom foot. Later: carry a light bag of groceries in the hand that isn't on the rail.

Safety callouts: If stairs make you dizzy or you have fallen on stairs, do toe taps only until you've practiced balance for two weeks.

3:35–4:55 • Row

Time	Who	Spoken	On screen
3:35	Chang	Sit or stand with the chair behind you. Check the band first.	Row
3:50	Chang	Pull your elbows back past your ribs.	Easier: Closer to the door, or towel
4:03	Chang	Elbows close to your body.	

Time	Who	Spoken	On screen
4:11	Chang	One... two... breathe out... three... slow down... four.	
4:16	Chang	Tall chest, shoulders down.	
4:29	Chang	Breathe out as you pull.	

Cues: Elbows close to your body. Tall chest, shoulders down. **Easier:** Sit closer to the door so the band is slacker, or do the towel pull-apart.

Harder: Scoot back for more tension, or pause 2 seconds with elbows back.

Safety callouts: Check the band for nicks before every use. Door must open away from you and be locked. Osteoporosis: keep the hinge small and the back long.

4:55–5:55 • Band press-out

Time	Who	Spoken	On screen
4:55	Chang	Sit or stand with the chair behind you. Check the band first.	Band press-out
5:07	Chang	Press your hands straight out in front of your chest.	Easier: Seated, closer to the door
5:17	Chang	Stay square. The band pulls, you don't turn.	
5:22	Chang	One... two... breathe out... three... slow down... four.	
5:26	Chang	Ribs down.	
5:36	Chang	Breathe out as you press.	

Cues: Stay square. The band pulls, you don't turn. Ribs down. **Easier:** Do it seated, or stand closer to the door. **Harder:** Step farther from the door, or hold 5 seconds.

Safety callouts: Door must open away from you and be locked.

5:55–7:10 • Two-hand carry

Time	Who	Spoken	On screen
5:55	Chang	Clear path. Walk along the counter or a hallway wall.	Two-hand carry
6:10	Chang	Stand tall, shoulders down.	Easier: One light bag, hand on counter
6:22	Chang	Tall, like you're proud of the groceries.	
6:34	Chang	Short, steady steps.	
6:46	Chang	Breathe steadily. You should be able to talk.	

Cues: Tall, like you're proud of the groceries. Short, steady steps. **Easier:** One light bag, and keep your free hand trailing along the counter. **Harder:** Heavier: two full gallon jugs (about 8 pounds each). Iron track: two heavier bags or kettlebells.

Safety callouts: Hand arthritis flaring today? Squeeze gently, or skip and do the finger spread only. Pick up and put down with a hinge. Never round your back to lift.

7:10–8:05 • Curls

Time	Who	Spoken	On screen
7:10	Chang	Sit tall. Chair against the wall.	Curls

Time	Who	Spoken	On screen
7:21	Chang	Bend your elbows and bring the bottles toward your shoulders.	Easier: One arm, or empty bottles
7:30	Chang	Elbows stay by your sides.	
7:34	Chang	One... two... breathe out... three... slow down... four.	
7:39	Chang	No swinging.	
7:48	Chang	Breathe out as you curl up.	

Cues: Elbows stay by your sides. No swinging. **Easier:** One arm at a time, or empty bottles. **Harder:** Heavier: a half-gallon jug (about 4 pounds), then a full gallon jug (about 8 pounds), one at a time.

8:05–8:45 • Calf stretch

Time	Who	Spoken	On screen
8:05	Chang	Hand on the counter. Chair behind you.	Calf stretch
8:14	Chang	Bend the front knee and lean gently toward the wall until you feel a stretch in the back calf.	Easier: Smaller step back
8:20	Chang	Back heel stays down.	
8:27	Chang	Toes point forward.	
8:33	Chang	Breathe slowly through the stretch.	

Cues: Back heel stays down. Toes point forward. **Easier:** Smaller step back. **Harder:** Bigger step back, or bend the back knee slightly to reach the lower calf.

8:45–9:25 • Cyclic sighing

Time	Who	Spoken	On screen
8:45	Chang	Sit. Two sniffs in through the nose. One long sigh out.	Cyclic sighing
9:04	Chang	No holding.	Easier: Just slow nose breathing

Cues: Two sniffs in, one long sigh out. No holding at any point. **Easier:** Just breathe slowly through your nose, making the out-breath a little longer. **Harder:** Build up to 5 minutes.

Safety callouts: If you feel light-headed, return to normal breathing.

9:25–9:50 • Close

Time	Who	Spoken	On screen
9:25	Chang	Strong at 70. Not a slogan. A checkpoint.	
9:38	Chang	Your final checkpoint is this weekend.	

Where the numbers come from

Every number and health statement in this book traces to one of the entries below. They are the same evidence IDs our whole team uses (EVIDENCE.md). Grade A means a Cochrane review, a large meta-analysis or a guideline body. B means a single good trial or a small meta-analysis. C means an observational study: it shows a link, not cause and effect. D means lab, physiology or label data.

E01 (grade A). Progressive resistance training in older adults: 121 RCTs, 6,700 people, 2–3×/week. Large effect on strength (SMD 0.84), moderate-to-large effect on getting out of a chair (SMD –0.94), gait speed +0.08 m/s, less osteoarthritis pain. Most side effects were minor soreness or joint pain.

Source: Liu & Latham, Cochrane 2009 — https://www.cochrane.org/evidence/CD002759_progressive-resistance-strength-training-improving-physical-function-older-adults

E02 (grade A). NSCA position statement: resistance training is safe and effective for older adults, including frail people. Progress gradually. Train 2–3×/week, 1–3 sets of 6–12 reps at 70–85% 1RM once adapted. Include power (fast lifting intent) and balance.

Source: Fragala et al., J Strength Cond Res 2019 — https://journals.lww.com/nsca-jscr/fulltext/2019/08000/resistance_training_for_older_adults__position.1.aspx

E03 (grade B (small)). Ten frail nursing-home residents aged ~90, 8 weeks of high-intensity leg training: strength +174%, thigh muscle area up about 9%, walking speed improved. Muscle still adapts in the 90s.

Source: Fiatarone et al., JAMA 1990 — <https://pubmed.ncbi.nlm.nih.gov/2342214/>

E04 (grade A (review)). Around age 75, muscle mass drops about 0.6–1% per year while strength drops 2.5–4% per year. Strength declines 2–5× faster than mass.

Source: Mitchell et al., *Frontiers in Physiology* 2012 — <https://www.frontiersin.org/journals/physiology/articles/10.3389/fphys.2012.00260/full>

E06 (grade A (consensus)). Sarcopenia screening cut-offs (EWGSOP2): grip under 27 kg for men or under 16 kg for women, or 5 chair rises taking more than 15 s, signals "probable sarcopenia".

Source: Cruz-Jentoft et al., *Age Ageing* 2019 — <https://www.esceo.org/sites/esceo/files/pdf/2019%20Age%20Ageing%20EWGSOP2.pdf>

E08 (grade C). Sitting-rising test (sit to the floor and stand up without support; score 0–10, lose 1 point per hand/knee support). 2,002 adults aged 51–80. The lowest scorers (0–2) had 5–6× the death risk of the highest (8–10). Each +1 point was linked with 21% lower mortality.

Source: Brito, Araújo et al., *Eur J Prev Cardiol* 2012/2014 — <https://www.sciencedaily.com/releases/2012/12/121213085202.htm>

E11 (grade A (tool)). CDC STEADI 30-second chair stand (17-inch chair, arms crossed), used in the CDC's older-adult screening. Below-average scores: men 60–64 <14, 65–69 <12, 70–74 <12, 75–79 <11, 80–84 <10, 85–89 <8, 90–94 <7. Women 60–64 <12, 65–69 <11, 70–74 <10, 75–79 <10, 80–84 <9, 85–89 <8, 90–94 <4. If arms are used to stand, the score is 0.

Source: CDC STEADI — <https://www.cdc.gov/steady/media/pdfs/STEADI-Assessment-30Sec-508.pdf>

E39 (grade A–). High-velocity "power" training (lift with fast intent, lower slowly) gives functional gains similar to or slightly better than traditional training in older adults, and it is safe when supervised and progressed.

Source: J Physiotherapy 2023 systematic review — <https://www.sciencedirect.com/science/article/pii/S183695532300053X> ; Eur Rev Aging Phys Act 2022 — <https://link.springer.com/article/10.1186/s11556-022-00297-x>

E40 (grade A (consensus)). Osteoporosis exercise guidance: do resistance and balance training. Avoid rapid, repeated or loaded spinal flexion (sit-ups, toe-touches with weight) and forceful twisting. Flexion-exercise programs were linked with more vertebral fractures.

Source: Giangregorio et al., "Too Fit To Fracture", Osteoporos Int 2014 — <https://link.springer.com/10.1007/s00198-014-2881-4> ; Sinaki & Mikkelsen, Arch Phys Med Rehabil 1984

E41 (grade B (known literature)). Pain-monitoring model: during and after exercise, pain up to about 3/10 (we use a conservative threshold) is acceptable if it settles by the next morning and isn't trending up week to week.

Source: Thomeé, Phys Ther 1997; Silbernagel et al., Am J Sports Med 2007

E42 (grade A (statement)). AHA scientific statement (2023): resistance training is recommended for adults with and without cardiovascular disease. It lowers blood pressure and improves glucose control and function. Avoid breath-holding (Valsalva). People with CVD should get individualized clearance.

Source: Paluch et al., Circulation 2024 — <https://www.ahajournals.org/doi/10.1161/CIR.0000000000001189>

E43 (grade A (guideline)). Pre-exercise screening (ACSM 2015): anyone with symptoms of cardiovascular, metabolic or renal disease (chest pain, unusual breathlessness, fainting) should get medical clearance before starting. Otherwise, light-to-moderate exercise can begin and progress.

Source: Riebe et al., Med Sci Sports Exerc 2015 (known literature)

E45 (grade B (known literature)). Getting-up-from-the-floor strategies can be taught to older adults. Practice it before you ever need it (backward chaining: kneel → half-kneel → stand, using sturdy furniture).

Source: Hofmeyer et al., J Am Geriatr Soc 2002 (floor-rise strategy training)

E49 (grade A (tool/norms)). Senior Fitness Test (Rikli & Jones): 7,183 US community-dwelling adults aged 60–94 from 267 sites in 21 states. "Normal range" = the middle 50% (25th–75th percentile) for each 5-year age band and sex. 30-s chair stand, 30-s arm curl (5 lb women / 8 lb men), 2-minute step test (right-knee lifts to midway between kneecap and hip bone), chair sit-and-reach (inches, + past the toes / – short of them). Example normal ranges, women 70–74: chair stand 10–15, arm curl 12–17, step test 68–101, sit-and-reach –1.0 to +4.0 in; men 70–74: 12–17, 14–21, 80–110, –3.5 to +2.5 in. Full tables in products/strength_reset.md. It's a fitness comparison, not a diagnosis.

Source: Rikli & Jones, Senior Fitness Test Manual, 2nd ed. 2013 (primary not opened).

Opened: normative tables — <https://integrativephysicaltherapyservices.com/wp-content/uploads/2023/08/Fitness-Testing-for-Seniors-PDF.pdf> ; protocol + sample size — <https://fitnessnorms.com/functional/chair-sit-and-reach/> , <https://fitnessnorms.com/functional/step-test/> , <https://fitnessnorms.com/functional/arm-curl/> . Note: the PDF's women 60–64 sit-and-reach cell reads "–0.5 to +0.5" (typo); we use the fitnessnorms 25th–75th percentiles (–0.6 to +4.8 in).